Example from two weeks ago

$$A = \begin{bmatrix} \frac{1+i}{2} & 1 \\ 0 & 1 \end{bmatrix}. \quad \text{Eigenvalues } \frac{1+i}{2} \text{ and } 1.$$

$$\frac{1+i}{2}\text{-eigenspace} = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

$$1\text{-eigenspace} = \operatorname{span}\left\{ \begin{bmatrix} 1+i \\ 1 \end{bmatrix} \right\}$$

Set $P = \begin{bmatrix} 1 & 1+i \\ 0 & 1 \end{bmatrix}$. Then $A = P \underbrace{\begin{bmatrix} \frac{1+i}{2} & 0 \\ 0 & 1 \end{bmatrix}}_{= D} P^{-1}$.

Exercise
$$\begin{aligned}
(PDP^{-1})^2 &= (PDP^{-1})(PDP^{-1}) \\
&= PD\cancel{P^{-1}}\cancel{P}DP^{-1} \\
&= PDIDP^{-1} \\
&= PDD\,P^{-1} \quad (I \text{ does nothing under multiplication}) \\
&= PD^2P^{-1}
\end{aligned}$$

Similarly $(PDP^{-1})^3 = \cdots = PD^3P^{-1}$

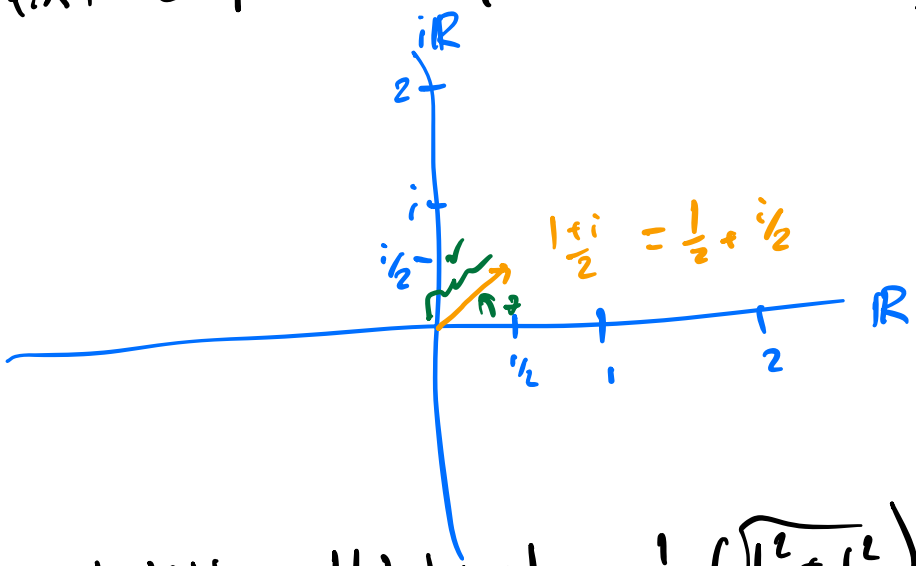
By induction $(PDP^{-1})^k = \cdots = PD^kP^{-1}$.

$$\Rightarrow \quad A^k = (PDP^{-1})^k = PD^kP^{-1}.$$

Now, $D^k = \begin{bmatrix} \frac{1+i}{2} & 0 \\ 0 & 1 \end{bmatrix}^k = \cdots = \begin{bmatrix} \left(\frac{1+i}{2}\right)^k & 0 \\ 0 & 1^k \end{bmatrix} = \begin{bmatrix} \left(\frac{1+i}{2}\right)^k & 0 \\ 0 & 1 \end{bmatrix}$.

Special properties of diagonal matrices.

We would like to compute $\left(\frac{1+i}{2}\right)^k$. For this we first compute the polar form of $\frac{1+i}{2}$.



$$r = \left|\frac{1+i}{2}\right| = \left|\frac{1}{2}\right| \, |1+i| = \frac{1}{2}\left(\sqrt{1^2+1^2}\right) = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} < 1$$

$$\theta = \pi/4$$

$$\Rightarrow \quad \frac{1+i}{2} = re^{i\theta} = \frac{1}{\sqrt{2}} e^{i\pi/4}$$

$$\Rightarrow \quad \left(\frac{1+i}{2}\right)^k = \left(\frac{1}{\sqrt{2}} e^{i\pi/4}\right)^k = \left(\frac{1}{\sqrt{2}}\right)^k e^{ik\pi/4}$$

As $k \to \infty$ $\left(\frac{1}{\sqrt{2}}\right)^k \to 0$ since $\frac{1}{\sqrt{2}} < 1$.

as $k \to \infty$ $e^{ik\pi/4}$ does not converge, but is bounded by $|e^{ik\pi/4}| = 1$.

$$\Rightarrow \quad \left(\frac{1+i}{2}\right)^k \to 0 \quad \text{as } k \to \infty$$

In conclusion $A^k \to P \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} P^{-1}$

as $k \to \infty$.

As k gets large A^k "kills off the $|\tau_i|$-eigenspaces, and fixes the 1-eigenspace.

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§6.1 Inner product, length, orthogonality.

Recall the definition of the dot product on $\mathbb{R}^n$.

Def Suppose $u = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \in \mathbb{R}^n$ and $v = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$.

Then the (standard) <u>inner product</u> (<u>dot product</u>) on $\mathbb{R}^n$ is definition.

$$\langle u, v \rangle = u \cdot v = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$
$$= \sum_{j=1}^{n} u_j v_j .$$

$$\left\langle \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\rangle = (1)(-1) + (2)(0) + (3)(1)$$
$$= -1 + 0 + 3$$
$$= 2.$$

Remarks - The inner product on $\mathbb{R}^n$ gives us information about the <u>length</u> or <u>norm</u> of a vector.

$$\|v\| = \sqrt{\langle v,v\rangle} = \sqrt{v_1^2+v_2^2+\cdots+v_n^2}$$

$$= \sqrt{\sum_{j=1}^{n} v_j^2} \geq 0.$$

- The inner product also gives us information about the angle between two vectors.

- $\langle \cdot,\cdot \rangle : \underbrace{\mathbb{R}^n \times \mathbb{R}^n}_{\text{pairs of vectors}} \to \underbrace{\mathbb{R}}_{\text{real numbers}}$ is a function.

Theorem 1

Suppose $u,v,w \in \mathbb{R}^n$. Then.

(a) $\langle u,v\rangle = \langle v,u\rangle$

(b) $\langle u+v,w\rangle = \langle u,w\rangle + \langle v,w\rangle$

(c) $\langle cu,w\rangle = c\langle u,w\rangle$ $\wedge$ $\langle u,cw\rangle = c\langle u,w\rangle$ for all $c \in \mathbb{R}$

$\langle \cdot,\cdot \rangle$ is linear in "first slot".

(d) $\langle v,v\rangle \geq 0$ and $\langle v,v\rangle = 0 \iff v=0$

(e) $\|cu\| = |c|\,\|u\|$ for all $c \in \mathbb{R}$.

We can imitate the inner product on $\mathbb{R}^n$ to get an inner product on the <u>complex</u> vector space $\mathbb{C}^n$.

Problem

Consider $\begin{bmatrix} 1+i \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \in \mathbb{C}^2$.

$$(1+i)(-1) + (1)(1) = (-1-i) + 1 = -i$$

Seems OK, but

$$\sqrt{(1+i)(1+i) + (1)(1)} = \sqrt{2i+1} = \sqrt{1+2i}$$

This is not a nonnegative real number, so doesn't appear to give us a good idea of norm or length. To get around this problem we define the inner product as follows.

Def. Suppose $u = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \in \mathbb{C}^n$ and $v = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{C}^n$.

The <u>inner product</u> of u and v is

$$\langle u, v \rangle = u_1 \overline{v_1} + u_2 \overline{v_2} + \dots + u_n \overline{v_n}$$

complex conjugates

This defines the inner product on $\mathbb{C}^n$.

Example

$$\left\langle \begin{bmatrix} 1+i \\ 1 \end{bmatrix}, \begin{bmatrix} 1+i \\ 1 \end{bmatrix} \right\rangle = (1+i)(\overline{1+i}) + (1)(\overline{1})$$
$$= (1+i)(1-i) + 1$$
$$= (1^2 + 1^2) + 1$$
$$= 3 \geq 0.$$

$$\sqrt{\left\langle \begin{bmatrix} 1+i \\ 1 \end{bmatrix}, \begin{bmatrix} 1+i \\ 1 \end{bmatrix} \right\rangle} = \sqrt{3}.$$

This tells us that $\begin{bmatrix} 1+i \\ 1 \end{bmatrix}$ has norm $\sqrt{3}$

with the following definition

Def. Suppose $v \in \mathbb{C}^n$. Then the <u>norm</u> (or <u>length</u>) of v is $\|v\| = \sqrt{\langle v, v \rangle} \geq 0$.

Remarks - WARNING. The inner product for $\mathbb{C}^n$ is <u>not</u> in the textbook.

- This inner product on $\mathbb{C}^n$ still gives us an idea of length, but the angle between two complex vector is not preserved, but the notion of "orthogonality" is preserved.

- WARNING - Don't forget the complex conjugation in the inner product.

Theorem

Suppose $u, v, w \in \mathbb{C}^n$. Then

(a) $\langle u, v \rangle = \overline{\langle v, u \rangle}$

(b) $\langle u+v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$

(c) $\langle cu, v \rangle = c \langle u, v \rangle$, $\langle u, cv \rangle = \bar{c} \langle u, v \rangle$ for all $c \in \mathbb{C}$.

(d) $\langle v, v \rangle \geq 0$ and $\langle v, v \rangle = 0 \iff v = 0$

(e) $\|cv\| = |c| \, \|v\|$ for all $c \in \mathbb{C}$.

Sketch of proof (c)

Take $u = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$, $v = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$, $c \in \mathbb{C}$. Then

$$\begin{aligned}
\langle u, cv \rangle &= u_1 \overline{cv_1} + u_2 \overline{cv_2} + \cdots + u_n \overline{cv_n} \\
&= u_1 \bar{c} \bar{v}_1 + u_2 \bar{c} \bar{v}_2 + \cdots + u_n \bar{c} \bar{v}_n \\
&= \bar{c} \left(u_1 \bar{v}_1 + \cdots + u_n \bar{v}_n \right) \\
&= \bar{c} \langle u, v \rangle.
\end{aligned}$$